

RHYL, United Kingdom

WMO: 033130

Lat: 53.2592N

Lon: 3.5089W

Elev: 249

StdP: 14.56

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.7	30.1	21.8	16.5	31.7	24.2	18.5	33.2	28.2	47.6	25.7	47.9	5.1	210	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	74.4	63.7	71.1	62.4	68.4	61.1	65.4	72.0	63.7	69.3	62.1	67.0	7.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	86.2	68.3	61.2	81.3	66.6	59.8	77.5	65.2	30.3	71.9	29.1	69.4	28.0	67.2	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	19.9	17.6	DB	23.9	80.4	3.1	3.6	21.7	83.0	19.8	85.1	18.1	87.1	15.8	89.8	(4)
(5)			WB	22.9	69.8	3.4	3.8	20.5	72.5	18.5	74.7	16.6	76.9	14.1	79.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	41.8	42.1	44.2	47.6	52.3	57.3	60.5	60.3	56.9	52.2	46.6	42.7	(6)
(7)		DBStd	8.33	5.53	5.32	5.10	4.93	4.76	4.08	3.95	3.68	3.61	4.71	5.02	6.09	(7)
(8)		HDD50	1188	256	225	188	103	27	1	0	0	1	31	124	232	(8)
(9)		HDD65	5345	719	642	644	523	393	235	151	155	245	396	552	690	(9)
(10)		CDD50	1341	2	4	9	30	99	219	325	318	207	99	22	7	(10)
(11)		CDD65	23	0	0	0	0	0	3	11	8	1	0	0	0	(11)
(12)		CDH74	133	0	0	0	0	6	24	61	39	3	0	0	0	(12)
(13)		CDH80	17	0	0	0	0	0	2	10	5	0	0	0	0	(13)
(14)	Wind	WSAvg	7.9	9.6	9.5	9.1	7.7	7.0	6.8	6.6	6.7	6.7	7.6	8.2	9.0	(14)
(15)	Precipitation	PrecAvg	31.7	3.0	2.4	1.8	1.9	2.2	2.2	2.4	2.7	2.7	3.4	3.5	3.5	(15)
(16)		PrecMax	43.3	5.3	5.4	4.1	3.9	4.3	5.4	5.0	5.6	5.6	6.4	8.5	7.8	(16)
(17)		PrecMin	22.7	0.0	0.6	0.3	0.1	0.7	0.5	0.6	0.5	0.2	0.8	1.3	1.1	(17)
(18)		PrecStd	4.9	1.5	1.3	1.0	1.0	0.9	1.3	1.1	1.1	1.3	1.5	1.7	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	57.3	60.7	68.7	73.5	77.4	80.5	79.3	72.7	66.3	59.9	57.2	(19)
(20)			MCWB	51.9	51.3	51.9	56.8	62.6	64.7	66.0	65.9	63.4	59.9	56.1	53.5	(20)
(21)		2%	DB	53.3	53.6	56.1	61.3	67.6	71.7	74.5	72.3	68.5	63.1	57.8	55.1	(21)
(22)			MCWB	50.0	49.8	50.3	53.3	59.0	62.3	63.8	63.9	61.0	58.2	54.6	52.3	(22)
(23)		5%	DB	51.6	51.8	53.7	57.8	63.8	67.7	71.0	68.9	65.6	61.4	56.0	53.3	(23)
(24)			MCWB	48.5	48.2	49.2	51.3	56.6	59.9	62.4	62.2	59.6	56.9	52.8	50.3	(24)
(25)		10%	DB	50.0	50.1	51.7	55.3	60.6	64.8	68.0	66.6	63.4	59.6	54.2	51.4	(25)
(26)			MCWB	46.9	46.5	47.7	50.0	54.5	58.4	61.2	60.8	58.2	55.5	51.0	48.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	52.2	53.7	57.2	64.3	66.8	68.0	67.4	65.1	61.1	57.0	54.1	(27)
(28)			MCD B	55.0	55.4	58.3	66.3	72.9	73.6	75.7	75.1	70.6	64.5	59.4	56.4	(28)
(29)		2%	WB	50.5	50.3	51.2	54.4	60.1	63.0	65.3	64.9	62.5	59.3	55.2	52.6	(29)
(30)			MCD B	52.9	53.2	54.9	60.2	66.3	69.6	72.5	70.7	66.8	62.7	57.5	54.9	(30)
(31)		5%	WB	48.9	48.7	49.5	52.3	57.5	60.9	63.4	63.4	60.5	57.6	53.2	50.8	(31)
(32)			MCD B	51.3	51.6	53.2	56.9	62.4	66.4	69.5	68.0	64.4	60.9	55.7	53.2	(32)
(33)		10%	WB	47.3	46.9	48.1	50.4	55.3	59.1	61.8	61.9	58.9	55.8	51.3	48.7	(33)
(34)			MCD B	49.8	49.9	51.5	54.6	59.6	64.2	67.1	65.8	62.6	59.0	53.7	51.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	8.9	10.4	12.0	13.0	12.7	12.4	11.7	11.6	10.3	8.7	8.5	(35)
(36)			MCDBR	9.5	10.5	12.8	15.7	18.2	18.0	17.8	15.3	15.2	11.7	10.1	9.5	(36)
(37)			MCWBR	9.2	8.8	9.1	10.0	11.6	11.2	10.1	9.1	9.8	8.5	8.7	8.8	(37)
(38)		5% WB	MCDBR	9.5	10.1	11.8	14.4	16.7	16.0	15.6	13.6	13.0	10.8	9.8	9.7	(38)
(39)			MCWBR	9.5	9.0	9.0	9.9	11.3	10.9	10.1	8.7	9.6	8.9	9.0	9.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.342	0.373	0.395	0.400	0.393	0.394	0.378	0.368	0.351	0.327	0.319	(40)	
(41)		taud	2.299	2.356	2.288	2.253	2.291	2.340	2.345	2.401	2.424	2.423	2.363	2.273	(41)	
(42)		Ebn at Noon	200	239	255	264	269	272	270	268	256	233	201	178	(42)	
(43)		Edh at Noon	20	25	33	38	39	37	37	33	29	23	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	206	420	786	1215	1539	1617	1525	1252	903	502	250	156	(44)	
(45)		RadStd	20	46	69	115	107	153	153	84	68	43	26	16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page